

Project Specification & Ideation Document

Date:	1 Oct 2026	Team ID:	6ab63f444173fc565b5466f9
Project Name:	Auto Ticket Classification using Flow Designer	Maximum Marks:	4 Marks

IDEATION PHASE – PROBLEM STATEMENTS

Define the Problem Statements: Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love. A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face.

Customer Persona Example I am: a service desk user I'm trying to: submit IT incident tickets But: ticket categorization takes time Because: manual triage is required Which makes me feel: Frustrated with delayed routing	Target Operational Domain Domain: IT Service Management (ITSM) Environment: Enterprise Service Desk Automation Engine: Flow Designer Workflow Primary Goal: Automated zero-touch classification
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Problem Statement Matrix (PS)

ID	I am...	I'm trying to...	But...	Because...	Which makes me feel...
PS-1	Service Desk Team	classify incoming tickets automatically upon creation	manual ticket triage consumes excessive staff time	tickets require manual review of multiple categories	Delayed routing & backlogs
PS-2	Service Desk Team	route tickets to correct support groups promptly	wrong category assignments delay resolution	manual triage causes inconsistent classification	Inefficient handling

BRAINSTORMING & IDEA PRIORITIZATION

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process. Prioritizing volume over value, out-of-the-box ideas are welcome, leading to effective automation workflows.

Step 1: Problem Selection Selected Problem: Automatic ticket classification and intelligent group assignment using Flow Designer.	Step 2: Idea Listing Explored Concepts: Keyword matching, conditional branching, priority escalation, and automated notification triggers.	Step 3: Prioritization Selected Solution: High Importance / High Feasibility automated Flow Designer workflow engine.
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1. Solution Architecture Overview

Solution architecture bridges the gap between business problems and technology solutions. Its primary goals are:

- **Find the best tech solution** to solve existing business problems efficiently.
- **Describe the structure, characteristics, behavior,** and other key aspects of the software to project stakeholders.
- **Define features, development phases,** and specific solution requirements.
- **Provide technical specifications** according to which the solution is defined, managed, and delivered.

2. End-to-End Solution Architecture Diagram

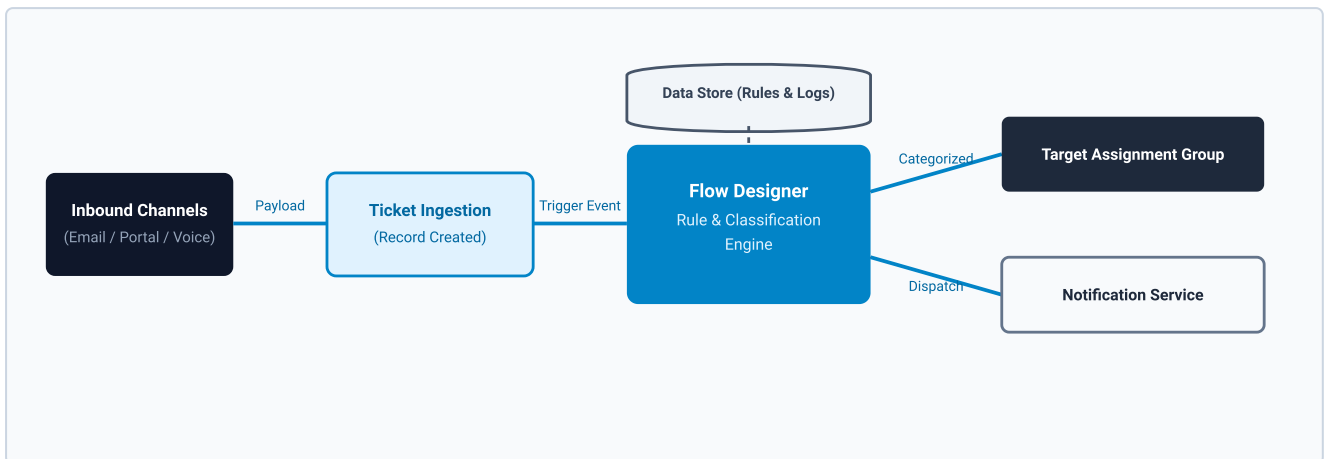
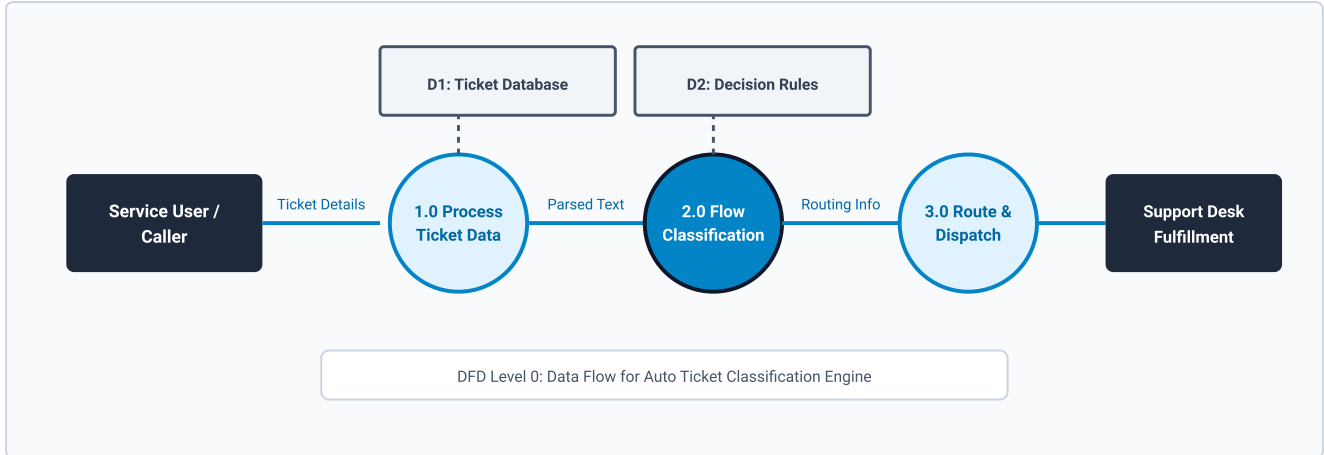


Figure 1: Architectural framework showing end-to-end classification and assignment flow for inbound IT tickets.

PROJECT DESIGN PHASE-II – DATA FLOW DIAGRAM (DFD)

A Data Flow Diagram (DFD) depicts system requirements graphically by illustrating how data enters and leaves the system, what processes transform the information, and where data is stored.

DFD Level 0 (Industry Standard Context Diagram)



User Stories Matrix

User Type	Epic	ID	User Story / Task	Acceptance Criteria	Prio	Rel
End User	Submission	USN-1	As a user, I submit an incident ticket with short description.	Ticket ID generated	High	S-1
End User	Notification	USN-2	As a user, I receive confirmation once auto-classified.	Email with category	High	S-1
Agent	Auto Triage	USN-3	As an agent, I want tickets pre-classified into Category.	Category matches rule	High	S-1
Manager	Rules UI	USN-4	As a manager, I can configure Flow Designer rules visually.	Rule updated without code	Med	S-2

PROJECT DESIGN PHASE-II – TECHNOLOGY STACK (TABLE 1)

The technical architecture and component breakdown for the Auto Ticket Classification project specifies the hardware, software, services, and integration layers required for zero-touch ticket routing.

Table 1: Technical Stack Component Breakdown

S.No	Component	Description	Technology / Framework
1.	User Interface	Service Portal & Web UI through which users submit IT incidents and check ticket status.	HTML5, CSS3, JavaScript, ServiceNow Service Portal / React
2.	Application Logic-1	Flow Designer Trigger Engine that executes upon new ticket record creation.	ServiceNow Flow Designer, JavaScript Engine
3.	Application Logic-2	Keyword parsing and string evaluation logic for short description extraction.	ServiceNow Script Include / Regular Expressions Engine
4.	Application Logic-3	Automated assignment logic that maps parsed categories to support groups.	Flow Designer Conditional Branching Rules
5.	Database	Relational database holding ticket schemas, category definitions, and rule sets.	Relational ITSM DB (MySQL / MariaDB / PostgreSQL)
6.	Cloud Database	Managed cloud data platform hosting operational ticket records and logs.	ServiceNow Cloud Database / Cloud SQL
7.	File Storage	Storage for ticket attachments, screenshots, and system activity logs.	Cloud Object Storage / Local Filesystem Storage
8.	External API-1	Inbound REST Webhook API for third-party monitoring alerts to auto-create tickets.	REST Webhook / JSON API
9.	External API-2	Email Gateway API for processing incoming support emails into ticket records.	Inbound Email Actions / SMTP Gateway API
10.	Machine Learning Model	Text classification model for intent detection and category matching.	ServiceNow Predictive Intelligence / Python NLTK / TF-IDF
11.	Infrastructure	Enterprise cloud instance hosting Flow Designer engine and database.	ServiceNow Enterprise Cloud / Containerized Hosting

PROJECT DESIGN PHASE-II – TECHNOLOGY STACK (TABLE 2)

The table below details non-functional characteristics, security controls, performance considerations, and architectural parameters driving system implementation.

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology / Control
1.	Open-Source Frameworks	Open-source utility libraries and frameworks used for string parsing, JSON handling, and UI components.	Node.js, Express.js, Bootstrap 5, Python NLTK
2.	Security Implementations	Multi-layered security including Role-Based Access Control (RBAC), data encryption at rest and in transit, and TLS 1.3.	AES-256 Encryption, TLS 1.3, IAM Controls, OAuth 2.0, OWASP Rules
3.	Scalable Architecture	Microservices-oriented 3-tier architecture capable of handling high-volume concurrent ticket submissions during peak outages.	3-Tier Cloud Architecture, Asynchronous Flow Designer Execution
4.	Availability	High-availability cloud environment utilizing load balancers and redundant active-active database node clustering.	Cloud Load Balancer, Multi-AZ High Availability Node Clustering
5.	Performance	Sub-second classification execution time with fast query caching and optimized indexing on ticket decision tables.	Redis In-Memory Cache, Indexed DB Decision Tables, CDN

TECHNICAL ENVIRONMENT & DEPLOYMENT GUIDELINES

Infrastructural Demarcation • Local Environment: Developer Sandboxes & Unit Test Suites. • Cloud Infrastructure: Staging & Production Instances hosting Flow Designer Engine.	Integration & Interfaces • External Interfaces: REST Webhooks, Email Gateways. • Internal Interfaces: Predictive Intelligence ML models & Task Assignment tables.
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IMPLEMENTATION SUMMARY & VERIFICATION

1. Solution Scope & Deliverables

The Auto Ticket Classification using Flow Designer project replaces manual ticket categorization with an automated rule evaluation flow, achieving rapid incident triage, accurate group assignment, and complete audit logging.

Key Capabilities Delivered

- Automated trigger on ticket creation.
- Keyword & rule-based classification.
- Dynamic assignment group routing.
- Automated notification dispatch.
- Comprehensive execution logging.

Business Impact

- Reduction in manual triage hours.
- Faster mean-time-to-resolution (MTTR).
- Elimination of human misclassification errors.
- Standardized workflow auditing.

2. Final Compliance Verification Checklist

Specification Field	Project Value / Status	Verification
Team ID	6ab63f444173fc565b5466f9	Verified
Project Name	Auto Ticket Classification using Flow Designer	Verified
Date	1 Oct 2026	Verified
External Links / URLs	Completely Removed / Cleaned	Verified
Technology Stack Tables	Table-1 & Table-2 Fully Completed	Verified
Functional Requirements & DFD	Retained & Rendered in DFD Level 0	Verified